

GOVERNMENT SCHOOL MANAGEMENT SYSTEM

UI / UX Design Specification

Visual Identity, Design Tokens & Component Guidelines

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1. Design Philosophy

GSMS digitizes something every government school already runs on paper: the register — the attendance book, the mark sheet, the library ledger, the inventory log. Rather than dressing the product in generic dashboard chrome, the interface borrows its structure from that physical register: ruled rows, tabbed dividers, and a stamp-like confirmation mark. The goal is a system that a Principal or Librarian with limited digital experience recognizes instantly, while remaining fast and precise for power users like Administrators entering marks for an entire class.

This document defines the design tokens (color, type, spacing, motion), the core component library, the signature interaction pattern, and per-module screen guidance that implementers should follow when building the React/Tailwind frontend described in the Software Design Specification (GSMS-SDS-001).

1.1 Design Principles

- Structure is information — ruled rows and tabbed navigation mirror a real register, so layout itself explains what the data means.
- One confirmation pattern, everywhere — attendance, marks, returns and disbursements all resolve with the same stamp-mark, so users learn it once.
- Density with clarity — screens used daily (attendance, marks entry) favor compact, scannable rows over decorative whitespace.
- Quiet by default, precise on demand — the palette stays restrained; color is reserved for status (present/absent, issued/overdue, paid/pending).
- Accessible to the least technical user in the building — the Librarian and Parent roles are the design floor, not the Administrator.

2. Color System

The palette is drawn from the register itself: aged paper, blue-black ink, and the gold and green and red marks a teacher makes by hand. It is deliberately warmer and less corporate than a typical admin-panel blue-on-white theme.

	Token	Hex	Usage
	Paper	#F7F4EC	App background — replaces stark white with warm register paper.
	Ink Navy	#10233F	Primary text, headers, primary buttons, active nav tab.
	Ledger Blue	#35578C	Links, secondary actions, selected states, info badges.
	Seal Gold	#AD8A4E	Brand accent, key CTAs, the register confirmation stamp.
	Register Green	#3F6B4F	Positive status — present, issued, paid, on-time.
	Marking Red	#9C3B34	Alerts — absent, overdue, rejected, validation errors.
	Rule Grey	#C9C2B2	Ruled row dividers, borders, disabled states.

Usage discipline: Ink Navy and Paper carry the interface. Ledger Blue and Seal Gold are used sparingly for actions and confirmation. Register Green and Marking Red are reserved exclusively for status — never for decoration — so color always carries meaning a user can rely on.

3. Typography

Role	Typeface (web)	Used For
Display	Lora (serif)	Page titles, role-dashboard headers, report card headings — a printed, official register feel, used sparingly.
Body / UI	Inter (sans)	All body text, form labels, navigation, buttons, tables — the workhorse face for dense daily-use screens.
Data / Mono	IBM Plex Mono	Student & employee IDs, dates, timetable period codes, marks — fixed-width so ledger columns align.

3.1 Type Scale

Token	Size / Line-height	Weight	Example Use
display-lg	32px / 40px	Lora Medium	Dashboard page title, e.g. “Attendance — Grade 9B”
display-sm	22px / 28px	Lora Medium	Section headers, card group titles
body-lg	16px / 24px	Inter Regular	Primary reading text, form field values
body-md	14px / 20px	Inter Regular	Table cells, list items, secondary UI text
label-sm	12px / 16px	Inter SemiBold, uppercase, tracked	Form labels, table headers, status badges
mono-data	13px / 18px	IBM Plex Mono Regular	IDs, dates, marks, timetable codes

4. Layout & Grid

4.1 Application Shell

The shell has three regions: a fixed left **Register Spine** (primary navigation, styled as a stack of tabbed dividers — one tab per module), a top bar (school/role context, search, notifications), and the main content area, which renders each module's ruled-row content.

ASCII layout concept:

```
+--+-----+
|R | Top bar: School name · Role · Search · Bell · Avatar |
|e +-----+
|g | |
|i | Page title (Lora) |
|s | ----- |
|t | Ruled row ..... ● |
|e | Ruled row ..... ● |
|r | Ruled row ..... ● |
+--+-----+
```

4.2 Spacing & Grid Tokens

Token	Value	Use
space-1	4px	Icon-to-label gap, badge padding
space-2	8px	Compact row padding (dense tables)
space-3	12px	Standard field/control padding
space-4	16px	Card padding, form row gaps
space-6	24px	Section spacing
space-8	32px	Page-level top/bottom margins
radius-sm / radius-md	4px / 8px	Inputs & badges / cards & modals — kept small, never fully rounded, to preserve the ledger/document feel
grid	12-column, 24px gutter, max width 1280px	Content area grid on desktop; collapses to single column below 768px

5. Signature Element — The Register Mark

Every module eventually asks a user to confirm a record: a teacher marks attendance, submits exam marks, a librarian records a return, an administrator disburses a welfare item. GSMS unifies all of these under one recognizable interaction: the **Register Mark** — a small circular stamp that fills in with a hand-drawn-style tick when a row is confirmed, echoing the checkmark a teacher makes in a physical register.

- Unconfirmed row: an empty outlined circle sits at the row's end, next to a thin dotted leader rule.
- On confirmation (save/submit): the circle fills with Register Green (positive) or Marking Red (negative, e.g. absent) and a short tick-draw animation plays (180ms, ease-out).
- Bulk actions (e.g. marking a whole class present): marks fill in sequence, left to right, row by row, at 40ms intervals — fast enough not to slow the teacher down, visible enough to confirm the action registered.
- The same component is reused, unstyled beyond color, across Attendance, Examination, Library returns, Inventory checks and Welfare disbursement — so the meaning (“this record is settled”) stays constant everywhere it appears.

This is the one place the interface takes a visible design risk; everywhere else, motion and ornament stay quiet so the Register Mark remains distinct and memorable rather than one effect among many.

6. Navigation — The Register Spine

The left navigation renders each module as a tabbed divider, the width of the icon plus a short vertical label, echoing indexed dividers in a physical binder. The active module's tab extends slightly into the content area and switches to Ink Navy with Paper-colored text; inactive tabs sit flush against the spine in Rule Grey. Module order follows the daily rhythm of school administration rather than alphabetical order:

Order	Module	Rationale
1	Dashboard	Landing view, role-specific summary
2	Attendance	First task of the school day
3	Timetable	Referenced throughout the day
4	Academic / Examination	Periodic, heavier data-entry tasks
5	Student / Teacher / Staff	Records referenced across other modules
6	Library, Inventory, Welfare	Operational modules, grouped together
7	Communication	Announcements & notifications
8	Reports	End-of-flow, cross-module output
9	User Management	Administrative, least frequently used

7. Component Library

7.1 Buttons

Variant	Style	Use
Primary	Ink Navy fill, Paper text, radius-sm	Main action per screen (Save, Submit Marks, Publish Timetable)
Secondary	Ledger Blue outline, Ledger Blue text	Alternate actions (Cancel, Export, Filter)
Accent	Seal Gold fill, Ink Navy text	Rare, high-emphasis calls to action (e.g. "Publish Report Cards")
Destructive	Marking Red outline, fills solid on hover	Delete, de-register, cancel enrolment
Text button	No fill, Ledger Blue text, underline on hover	Inline/low-emphasis actions within table rows

7.2 Form Inputs

- Inputs use a single bottom rule by default (echoing a ruled form), converting to a full outline only on focus or validation error, to keep dense forms visually quiet.
- Labels sit above the field in label-sm (uppercase, tracked) so screen scanning by less technical users is easier than placeholder-only inputs.
- Inline validation appears directly beneath the field in Marking Red, phrased as a plain instruction ("Enter a date on or before today," not "Invalid input").
- Required fields are marked with a small Seal Gold dot rather than a red asterisk, reserving red exclusively for errors.

7.3 Data Tables (Register Rows)

- Rows are separated by a single Rule Grey hairline, not zebra striping, matching ruled paper; a light Paper-tint band appears only on hover/focus for scanability.
- The rightmost column in workflow tables (attendance, exam entry, library circulation) is reserved for the Register Mark described in Section 5.
- Numeric and ID columns are right-aligned in mono-data type so figures line up the way they would in a handwritten ledger.
- Tables support sticky headers and column sort; dense tables (attendance, marks entry) default to compact row height (space-2), all others to standard (space-3).

7.4 Status Badges

Badge	Color	Example
Present / Issued / Paid / Active	Register Green text on tinted Green background	Attendance, library, welfare, user status
Absent / Overdue / Rejected	Marking Red text on tinted Red background	Attendance, library, examination resubmission
Pending / In Review	Seal Gold text on tinted Gold background	Welfare enrolment, marks awaiting approval
Neutral / Informational	Ledger Blue text on tinted Blue background	Draft timetable, unpublished report card

7.5 Cards & Dashboard Tiles

Dashboard summary tiles use radius-md, a 1px Rule Grey border, and Paper background — never a drop shadow heavier than a 2px soft blur, keeping the flat, paper-like quality. Each tile pairs one mono-data figure (e.g. “94%” attendance) with a label-sm caption and, where relevant, a small trend indicator in Register Green or Marking Red.

7.6 Notifications & Empty States

- Toasts confirm in the interface's own voice and match the action verb used on the trigger: a “Publish” button produces “Timetable published.”
- Errors state what happened and how to fix it, without apologizing (“Couldn't save — two periods overlap on Tuesday. Adjust one and try again.”).
- Empty states (e.g. no exams scheduled yet) show a short instruction and the single primary action to resolve it, not decorative illustration.

8. Per-Module Screen Guidance

Module	Primary Screen	Key Design Notes
Dashboard	Role-specific summary	Principal sees school-wide tiles (attendance %, exam pass rate, welfare coverage); Teacher sees today's classes and pending marks entry; Parent sees child's attendance streak and latest results.
Attendance	Class roster, period view	Full roster as ruled rows with Register Mark per student; single tap/click cycles Present → Absent → Late; bulk "Mark all present" as the primary button.
Timetable	Weekly grid	Grid view with subject/teacher/room per cell; conflict cells highlight in Marking Red until resolved; publish action is Accent (Seal Gold).
Examination	Marks entry sheet	Spreadsheet-like entry with mono-data figures, tab-to-next-cell keyboard flow, and a running class average shown in the header while entering.
Library	Catalogue & circulation	Search-first layout; issue/return actions produce an immediate Register Mark; overdue items surface at the top of the librarian's queue in Marking Red.
Inventory	Stock ledger	Ledger-row layout identical in spirit to Attendance; low-stock rows get a persistent Gold badge until restocked.
Welfare	Program enrolment & disbursement	Eligibility shown as a checklist before enrolment; disbursement confirmation uses the Register Mark, consistent with Attendance and Library.
Communication	Announcement composer & inbox	Composer keeps audience selection (role/class) visible at all times above the message body to prevent mis-sent announcements.
Reports	Filterable report builder	Filters persist as chips above the result table; export buttons (PDF/Excel) are Secondary style, since export is common but not the primary action.

9. Responsive Behavior

Breakpoint	Behavior
≥ 1280px (desktop)	Full Register Spine visible with labels; 12-column content grid.
768–1279px (tablet)	Spine collapses to icon-only; content grid drops to 8 columns; tables gain horizontal scroll for wide data.
< 768px (mobile — Parent/Student priority)	Spine becomes a bottom tab bar of the 4 most-used items (Dashboard, Attendance, Timetable, Announcements); tables convert to stacked ruled cards, one record per card.

10. Accessibility

- All text/background pairs meet WCAG 2.1 AA contrast (Ink Navy on Paper exceeds AAA; status colors are paired with a text label, never color alone).
- Every interactive element has a visible keyboard focus ring in Ledger Blue, 2px offset.
- The Register Mark's color change is always accompanied by an icon change (empty circle → check / cross), so status is legible without color perception.

- Motion (tick-draw, row fills) respects prefers-reduced-motion, falling back to an instant state change.
- Minimum touch target size of 40×40px on tablet/mobile for all row-level actions.

11. Motion Guidelines

- Motion is functional, not ambient: used to confirm actions (Register Mark), reveal state changes (row expand), and orient navigation (tab switch slide, 150ms).
- No page-load animation sequences or decorative background motion — the register metaphor is static and documentary by nature.
- Standard easing: ease-out for entrances/confirmations (180–220ms), ease-in for exits/dismissals (120–150ms).